

2

Operators, measurement and time evolution

In the last chapter we saw that each quantum state of a system is represented by a point or ‘ket’ $|\psi\rangle$ that lies in an abstract vector space. We saw that states for which there is no uncertainty in the value that will be measured for an ‘observable’ such as energy, form a set of basis states for this space – these basis states are analogous to the unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ of ordinary vector geometry. In this chapter we develop these ideas further by showing how every observable such as position, momentum or energy is associated with an operator on state space. We shall see that the energy operator plays a special role in that it determines how a system’s ket $|\psi\rangle$ moves through state space over time. Using these operators we are able at the end of the chapter to study the dynamics of a free particle, and to understand how the uncertainties in the position and momentum of a particle are intimately connected with one another, and how they evolve in time.

2.1 Operators

A **linear operator** on the vector space V is an object Q that transforms kets into kets in a linear way. That is, if $|\psi\rangle$ is a ket, then $|\phi\rangle = Q|\psi\rangle$ is another ket, and if $|\chi\rangle$ is a third ket and α and β are complex numbers, we have

$$Q(\alpha|\psi\rangle + \beta|\chi\rangle) = \alpha(Q|\psi\rangle) + \beta(Q|\chi\rangle). \quad (2.1)$$

Consider now the linear operator

$$I = \sum_i |i\rangle\langle i|, \quad (2.2)$$

where $\{|i\rangle\}$ is any set of basis kets. I really is an operator because if we apply it to any ket $|\psi\rangle$, we get a linear combination of kets, which must itself be a ket:

$$I|\psi\rangle = \sum_i |i\rangle\langle i|\psi\rangle = \sum_i (\langle i|\psi\rangle) |i\rangle, \quad (2.3)$$

where we are able to move $\langle i|\psi\rangle$ around freely because it's just a complex number. To determine which ket $I|\psi\rangle$ is, we substitute into (2.3) the expansion (1.12) of $|\psi\rangle$ and use the orthogonality relation (1.17):

$$\begin{aligned} I|\psi\rangle &= \sum_i |i\rangle\langle i| \left(\sum_j a_j |j\rangle \right) \\ &= \sum_i a_i |i\rangle = |\psi\rangle. \end{aligned} \quad (2.4)$$

We have shown that I applied to an arbitrary ket $|\psi\rangle$ yields that same ket. Hence I is the **identity operator**. We shall make extensive use of this fact.

Consider now the operator

$$H = \sum_i E_i |E_i\rangle\langle E_i|. \quad (2.5)$$

This is the most important single operator in quantum mechanics. It is called the **Hamiltonian** in honour of W.R. Hamilton, who introduced its classical analogue.¹ We use H to operate on an arbitrary ket $|\psi\rangle$ to form the ket $H|\psi\rangle$, and then we bra through by the adjoint $\langle\psi|$ of $|\psi\rangle$. We have

$$\langle\psi|H|\psi\rangle = \sum_i E_i \langle\psi|E_i\rangle\langle E_i|\psi\rangle. \quad (2.6)$$

By equation (1.24) $\langle E_i|\psi\rangle = a_i$, while by (1.19) $\langle\psi|E_i\rangle = a_i^*$. Thus

$$\langle\psi|H|\psi\rangle = \sum_i E_i |a_i|^2 = \sum_i p_i E_i = \langle E \rangle. \quad (2.7)$$

Here is yet another result of fundamental importance: if we squeeze the Hamiltonian between a quantum state $|\psi\rangle$ and its adjoint bra, we obtain the expectation value of the energy for that state.

¹ William Rowan Hamilton (1805–1865) was a protestant Irishman who was appointed the Andrews' Professor of Astronomy at Trinity College Dublin while still an undergraduate. Although he did not contribute to astronomy, he made important contributions to optics and mechanics, and to pure mathematics with his invention of quaternions, the first non-commutative algebra.

It is straightforward to generalise this result for the expectation value of the energy to other measurable quantities: if Q is an observable (i.e., something that we can measure) and its spectrum of possible values is $\{q_i\}$, then we expand an arbitrary ket $|\psi\rangle$ as a linear combination of states $|q_i\rangle$ in which the value of Q is well defined,

$$|\psi\rangle = \sum_i a_i |q_i\rangle, \quad (2.8)$$

and with Q we associate the operator

$$Q = \sum_i q_i |q_i\rangle \langle q_i|. \quad (2.9)$$

Then $\langle\psi|Q|\psi\rangle$ is the expectation value of Q when our system is in the state $|\psi\rangle$. When the state in question is obvious from the context, we shall sometimes write the expectation value of Q simply as $\langle Q\rangle$.

When a linear operator R turns up in any mathematical problem, it is generally expedient to investigate its eigenvalues and eigenvectors. An **eigenvector** is a vector that R simply rescales, and its **eigenvalue** is the rescaling factor. Thus, let $|r\rangle$ be an eigenvector of R , and r be its eigenvalue, then we have

$$R|r\rangle = r|r\rangle. \quad (2.10)$$

What are the eigenvectors and eigenvalues of H ? If we apply H to $|E_k\rangle$, we find

$$H|E_k\rangle = \sum_i E_i |E_i\rangle \langle E_i|E_k\rangle = E_k |E_k\rangle. \quad (2.11)$$

So the eigenvectors of H are the states of well-defined energy, and its eigenvalues are the possible results of a measurement of energy. Clearly this important result generalises immediately to eigenvectors and eigenvalues of the operator Q that we have associated with an arbitrary observable.

Consider the complex number $\langle\phi|Q|\psi\rangle$, where $|\phi\rangle$ and $|\psi\rangle$ are two arbitrary quantum states. After expanding the states in terms of the eigenvectors of Q , we have

$$\langle\phi|Q|\psi\rangle = \left(\sum_i b_i^* \langle q_i| \right) Q \left(\sum_j a_j |q_j\rangle \right) = \sum_{ij} b_i^* a_j q_j \delta_{ij} = \sum_i b_i^* q_i a_i. \quad (2.12)$$

Similarly, $\langle\psi|Q|\phi\rangle = \sum_i a_i^* q_i b_i$. Hence so long as the spectrum $\{q_i\}$ of Q consists entirely of real numbers (which is physically reasonable), then

$$(\langle\phi|Q|\psi\rangle)^* = \langle\psi|Q|\phi\rangle \quad (2.13)$$

for any two states $|\phi\rangle$ and $|\psi\rangle$. An operator with this property is said to be **Hermitian**. Hermitian operators have nice properties. In particular, one can prove – see Box 2.1 – that they have real eigenvalues and mutually orthogonal eigenvectors, and it is because we require these properties on

Box 2.1: Hermitian operators

Let Q be a Hermitian operator with eigenvalues q_i and eigenvectors $|q_i\rangle$. Then we bra the defining equation of $|q_k\rangle$ through by $\langle q_i|$, and bra the defining equation of $|q_k\rangle$ through by $\langle q_i|$:

$$\langle q_k|Q|q_i\rangle = q_i\langle q_k|q_i\rangle \quad \langle q_i|Q|q_k\rangle = q_k\langle q_i|q_k\rangle.$$

We next take the complex conjugate of the second equation from the first. The left side then vanishes because Q is Hermitian, so with equation (1.22)

$$0 = (q_i - q_k^*)\langle q_k|q_i\rangle.$$

Setting $k = i$ we find that $q_i = q_i^*$ since $\langle q_i|q_i\rangle > 0$. Hence the eigenvalues are real. When $q_i \neq q_k$, we must have $\langle q_k|q_i\rangle = 0$, so the eigenvectors belonging to distinct eigenvalues are orthogonal.

physical grounds that the operators of observables turn out to be Hermitian. In Chapter 4 we shall find that Hermitian operators arise naturally from another physical point of view.

Although the operators associated with observables are always Hermitian, operators that are not Hermitian turn out to be extremely useful. With a non-Hermitian operator R we associate another operator $R^\dagger$ called its **Hermitian adjoint** by requiring that for any states $|\phi\rangle$ and $|\psi\rangle$ it is true that

$$(\langle\phi|R^\dagger|\psi\rangle)^* = \langle\psi|R|\phi\rangle. \quad (2.14)$$

Comparing this equation with equation (2.13) it is clear that a Hermitian operator Q is its own adjoint: $Q^\dagger = Q$.

By expanding the kets $|\phi\rangle$ and $|\psi\rangle$ in the equation $|\phi\rangle = R|\psi\rangle$ as sums of basis kets, we show that R is completely determined by the array of numbers (called **matrix elements**)

$$R_{ij} = \langle i|R|j\rangle. \quad (2.15)$$

In fact

$$\begin{aligned} |\phi\rangle &= \sum_i b_i|i\rangle = R|\psi\rangle = \sum_j a_j R|j\rangle \\ \Rightarrow b_i &= \sum_j a_j \langle i|R|j\rangle = \sum_j R_{ij}a_j. \end{aligned} \quad (2.16)$$

If in equation (2.14) we set $|\phi\rangle = |i\rangle$ and $|\psi\rangle = |j\rangle$, we discover the relation between the matrix of R and that of $R^\dagger$:

$$(R_{ij}^\dagger)^* = R_{ji} \quad \Leftrightarrow \quad R_{ij}^\dagger = R_{ji}^*. \quad (2.17)$$

Hence the matrix of $R^\dagger$ is the complex-conjugate transpose of the matrix for R . If R is Hermitian so that $R^\dagger = R$, the matrix R_{ij} must equal its complex-conjugate transpose, that is, it must be a Hermitian matrix.

Table 2.1 Rules for Hermitian adjoints

Object	i	$ \psi\rangle$	R	QR	$R \psi\rangle$	$\langle\phi R \psi\rangle$
Adjoint	-i	$\langle\psi $	$R^\dagger$	$R^\dagger Q^\dagger$	$\langle\psi R^\dagger$	$\langle\psi R^\dagger \phi\rangle$

Operators can be multiplied together: when the operator QR operates on $|\psi\rangle$, the result is what you get by operating first with R and then applying Q to $R|\psi\rangle$. We shall frequently need to find the Hermitian adjoints of such products. To find out how to do this we replace R in (2.17) by QR :

$$(QR)_{ij}^\dagger = (QR)_{ji}^* = \sum_k Q_{jk}^* R_{ki}^* = \sum_k R_{ik}^\dagger Q_{kj}^\dagger = (R^\dagger Q^\dagger)_{ij}. \quad (2.18)$$

Thus, to dagger a product we reverse the terms and dagger the individual operators. By induction it is now easy to show that

$$(ABC \dots Z)^\dagger = Z^\dagger \dots C^\dagger B^\dagger A^\dagger. \quad (2.19)$$

If we agree that the Hermitian adjoint of a complex number is its complex conjugate and that $|\psi\rangle^\dagger = \langle\psi|$ and $\langle\psi|^\dagger = |\psi\rangle$, then we can consider the basic rule (2.14) for taking the complex conjugate of a matrix element to be a generalisation of the rule we have derived about reversing the order and daggering the components of a product of operators. The rules for taking Hermitian adjoints are summarised in Table 2.1.

Functions of operators We shall frequently need to evaluate functions of operators. For example, the potential energy of a particle is a function $V(\hat{x})$ of the position operator $\hat{x}$. Let f be any function of one variable and R be any operator. Then we define the operator $f(R)$ by the equation

$$f(R) = \sum_i f(r_i) |r_i\rangle \langle r_i|, \quad (2.20)$$

where the r_i and $|r_i\rangle$ are the eigenvalues and eigenkets of R . This definition defines $f(R)$ to be the operator that has the same eigenkets as R and the eigenvalues that you get by evaluating the function f on the eigenvalues of R .

Commutators The commutator of two operators A, B is defined to be

$$[A, B] = AB - BA. \quad (2.21)$$

If $[A, B] \neq 0$, it is impossible to find a complete set of mutual eigenkets of A and B (Problem 2.19). Conversely, it can be shown that if $[A, B] = 0$ there is a complete set of mutual eigenkets of A and B , that is, there is a complete set of states of the system in which there is no uncertainty in the value that will be obtained for either A or B . We shall make extensive use of this fact. Notice that the word *complete* appears in both these statements; even in the case $[A, B] \neq 0$ it may be possible to find states in which both A and B have definite values. It is just that such states

cannot form a complete set. Similarly, when $[A, B] = 0$ there can be states for which A has a definite value but B does not. The literature is full of inaccurate statements about the implications of $[A, B]$ being zero or non-zero.

Three invaluable rules are

$$\begin{aligned}[A + B, C] &= [A, C] + [B, C] \\ AB &= BA + [A, B] \\ [AB, C] &= [A, C]B + A[B, C].\end{aligned}\tag{2.22}$$

All three rules are trivial to prove by explicitly writing out the contents of the square brackets. With these rules it is rarely necessary to write out the contents of a commutator again, so they eliminate a common source of error and tedium in calculations. Notice the similarity of the third rule to the standard rule for differentiating a product: $d(ab)/dc = (da/dc)b + a(db/dc)$. The rule is easily generalised by induction to the rule

$$[ABC \dots, Z] = [A, Z]BC \dots + A[B, Z]C \dots + AB[C, Z] \dots \tag{2.23}$$

We shall frequently need to evaluate the commutator of an operator A with a function f of an operator B . We assume that f has a convergent Taylor series² $f = f_0 + f'B + \frac{1}{2}f''B^2 + \dots$, where $f_0 = f(0)$, $f' = (df(x)/dx)_0$, etc., are numbers. Then

$$\begin{aligned}[A, f(B)] &= f'(0)[A, B] + \frac{1}{2}f''(0)([A, B]B + B[A, B]) \\ &\quad + \frac{1}{3!}f'''(0)([A, B]B^2 + B[A, B]B + B^2[A, B]) + \dots\end{aligned}\tag{2.24}$$

In the important case in which B commutes with $[A, B]$, this expression simplifies dramatically

$$[A, f(B)] = [A, B](f' + f''B + \frac{1}{2}f'''B^2 + \dots) = [A, B]\frac{df}{dB}.\tag{2.25}$$

We shall use this formula several times.

² If necessary, we expand $f(x)$ about some point $x_0 \neq 0$, i.e., in powers of $x - x_0$, so we don't need to worry that the series about the origin may not converge for all x .

2.2 Evolution in time

Since physics is about predicting the future, equations of motion lie at its heart. Newtonian dynamics is dominated by the equation of motion $\mathbf{f} = m\mathbf{a}$, where $\mathbf{f}$ is the force on a particle of mass m and $\mathbf{a}$ is the resulting acceleration. In quantum mechanics the analogous dynamical equation is the **time-dependent Schrödinger equation (TDSE)**:³

$$i\hbar \frac{\partial |\psi\rangle}{\partial t} = H|\psi\rangle. \quad (2.26)$$

For future reference we use the rules of Table 2.1 to derive from this equation the equation of motion of a bra:

$$-i\hbar \frac{\partial \langle\psi|}{\partial t} = \langle\psi|H, \quad (2.27)$$

where we have used the fact that H is Hermitian, so $H^\dagger = H$. The great importance of the Hamiltonian operator is due to its appearance in the TDSE, which must be satisfied by the ket of *any* system.

The TDSE is a statement about the physical world, so it can only be justified by experiment – we cannot hope to derive it mathematically. What we can, and must, do mathematically, is demonstrate that it is consistent with Newton’s laws for the motion of macroscopic bodies. This we shall do shortly. That done, we gradually gain conviction that the TDSE is the true quantum-mechanical equation of motion by working out its implications for harmonic oscillators, magnetically torqued electrons and hydrogen and helium atoms.

One perhaps surprising aspect of the TDSE we can justify straight-away: while Newton’s second law is a second-order differential equation, the TDSE is first order. Since it is first order, the boundary data at $t = 0$ required to solve for $|\psi, t\rangle$ at $t > 0$ comprise the ket $|\psi, 0\rangle$. If the equation were second order in time, like Newton’s law, the required boundary data would include $\partial|\psi\rangle/\partial t$. But $|\psi, 0\rangle$ by hypothesis constitutes a complete set of amplitudes; it embodies *everything* we know about the current state of the system. If mathematics required us to know something about the system in addition to $|\psi, 0\rangle$, then either $|\psi\rangle$ would not constitute a complete set of amplitudes, or physics could offer no hope of predicting the future, and it would be time to take up biology or accountancy, or whatever.

The TDSE tells us that states of well-defined energy evolve in time in an exceptionally simple way

$$i\hbar \frac{\partial |E_n\rangle}{\partial t} = H|E_n\rangle = E_n|E_n\rangle, \quad (2.28)$$

³ Beginners sometimes interpret the TDSE as stating that $H = i\hbar\partial/\partial t$. This is as unhelpful as interpreting $\mathbf{f} = m\mathbf{a}$ as a definition of $\mathbf{f}$. For Newton’s equation to be useful it has to be supplemented by a description of the forces acting on the particle. Similarly, the TDSE is useful only when we have another expression for H .

which implies that

$$|E_n, t\rangle = |E_n, 0\rangle e^{-iE_n t/\hbar}. \quad (2.29)$$

That is, the passage of time simply changes the phase of the ket at a rate $E_n/\hbar$.

We can use this result to calculate the time evolution of an arbitrary state $|\psi\rangle$. In the energy representation the state is

$$|\psi, t\rangle = \sum_n a_n(t) |E_n, t\rangle. \quad (2.30)$$

Substituting this expansion into the TDSE (2.26) we find

$$i\hbar \frac{\partial |\psi\rangle}{\partial t} = \sum_n i\hbar \left(\dot{a}_n |E_n\rangle + a_n \frac{\partial |E_n\rangle}{\partial t} \right) = \sum_n a_n H |E_n\rangle, \quad (2.31)$$

where a dot denotes differentiation with respect to time. The right side cancels with the second term in the middle, so we have $\dot{a}_n = 0$. Since the a_n are constant, on eliminating $|E_n, t\rangle$ between equations (2.29) and (2.30), we find that the evolution of $|\psi\rangle$ is simply given by

$$|\psi, t\rangle = \sum_n a_n e^{-iE_n t/\hbar} |E_n, 0\rangle. \quad (2.32)$$

We shall use this result time and again.

States of well-defined energy are unphysical and never occur in Nature because they are incapable of changing in any way, and hence it is impossible to get a system into such a state. But they play an extremely important role in quantum mechanics because they provide the almost trivial solution (2.32) to the governing equation of the theory, (2.26). Given the central role of these states, we spend much time solving their defining equation

$$H|E_n\rangle = E_n|E_n\rangle, \quad (2.33)$$

which is known as the **time-independent Schrödinger equation**, or **TISE** for short.

2.2.1 Evolution of expectation values

We have seen that $\langle\psi|Q|\psi\rangle$ is the expectation value of the observable Q when the system is in the state $|\psi\rangle$, and that expectation values provide a natural connection to classical physics, which is about situations in which the result of a measurement is almost certain to lie very close to the quantum-mechanical expectation value. We can use the TDSE to determine the rate of change of this expectation value:

$$\begin{aligned} i\hbar \frac{d}{dt} \langle\psi|Q|\psi\rangle &= -\langle\psi|HQ|\psi\rangle + i\hbar \langle\psi|\frac{\partial Q}{\partial t}|\psi\rangle + \langle\psi|QH|\psi\rangle \\ &= \langle\psi|[Q, H]|\psi\rangle + i\hbar \langle\psi|\frac{\partial Q}{\partial t}|\psi\rangle, \end{aligned} \quad (2.34)$$

where we have used both the TDSE (2.26) and its Hermitian adjoint (2.27) and the square bracket denotes a commutator – see (2.21). Usually operators are independent of time (i.e., $\partial Q/\partial t = 0$), and then the rate of change of an expectation value is the expectation value of the operator $-i[Q, H]/\hbar$. This important result is known as **Ehrenfest's theorem**.

If a time-independent operator Q happens to commute with the Hamiltonian, that is if $[Q, H] = 0$, then for any state $|\psi\rangle$ the expectation value of Q is constant in time, or a **conserved quantity**. Moreover, in these circumstances Q^2 also commutes with H , so $\langle\psi|(\Delta Q)^2|\psi\rangle = \langle Q^2\rangle - \langle Q\rangle^2$ is also constant. If initially ψ is a state of well-defined Q , i.e., $|\psi\rangle = |q_i\rangle$ for some i , then $\langle(\Delta Q)^2\rangle = 0$ at all times. Hence, whenever $[Q, H] = 0$, a state of well-defined Q evolves into another such state, so the value of Q can be known precisely at all times. The value q_i is then said to be a **good quantum number**. We always need to label states in some way. The label should be something that can be checked at any time and is not constantly changing. Good quantum numbers have precisely these properties, so they are much employed as labels of states.

If the system is in a state of well-defined energy, the expectation value of any time-independent operator is time-independent, even if the operator does not commute with H . This is true because in these circumstances equation (2.34) becomes

$$i\hbar \frac{d}{dt} \langle E|Q|E\rangle = \langle E|(QH - HQ)|E\rangle = (E - E)\langle E|Q|E\rangle = 0, \quad (2.35)$$

where we have used the equation $H|E\rangle = E|E\rangle$ and its Hermitian adjoint. In view of this property of having constant expectation values of all time-independent operators, states of well-defined energy are called **stationary states**.

Since H inevitably commutes with itself, equation (2.34) gives for the rate of change of the expectation of the energy

$$\frac{d\langle E\rangle}{dt} = \left\langle \frac{\partial H}{\partial t} \right\rangle. \quad (2.36)$$

In particular $\langle E\rangle$ is constant if the Hamiltonian is time-independent. This is a statement of the principle of the conservation of energy since time-dependence of the Hamiltonian arises only when some external force is working on the system. For example, a particle that is gyrating in a time-dependent magnetic field has a time-dependent Hamiltonian because work is being done either on or by the currents that generate the field.

2.3 The position representation

If the system consists of a single particle that can move in only one dimension, the amplitudes $\psi(x)$ to find the particle at x for x in $(-\infty, \infty)$ constitute a complete set of amplitudes. By analogy with equation (1.24) we have⁴

$$|\psi\rangle = \int_{-\infty}^{\infty} dx \psi(x)|x\rangle. \quad (2.37)$$

Here an integral replaces the sum because the spectrum of possible values of x is continuous rather than discrete. Our basis kets are the states $|x\rangle$ in which the particle is definitely at x . By analogy with equation (1.25) we have

$$\psi(x) = \langle x|\psi\rangle. \quad (2.38)$$

Notice that both sides of this equation are complex numbers that depend on the variable x , that is, they are complex-valued functions of x . For historical reasons, the function $\psi(x)$ is called the **wavefunction** of the particle. By the usual rule (1.22) for complex conjugation of a bra-ket we have

$$\psi^*(x) = \langle\psi|x\rangle. \quad (2.39)$$

The analogue for the kets $|x\rangle$ of the orthogonality relation (1.17) is

$$\langle x'|x\rangle = \delta(x - x'), \quad (2.40)$$

where the **Dirac delta function** $\delta(x - x')$ is zero for $x \neq x'$ because when the particle is at x , it has zero amplitude to be at a different location x' . We get insight into the value of $\delta(x - x')$ for $x = x'$ by multiplying equation (2.37) through by $\langle x'|$ and using equation (2.38) to eliminate $\langle x'|\psi\rangle$:

$$\begin{aligned} \langle x'|\psi\rangle &= \psi(x') = \int dx \psi(x) \langle x'|x\rangle \\ &= \int dx \psi(x) \delta(x - x'). \end{aligned} \quad (2.41)$$

Since $\delta(x - x')$ is zero for $x \neq x'$, we can replace $\psi(x)$ in the integrand by $\psi(x')$ and then take this number outside the integral sign and cancel it with the $\psi(x')$ on the left-hand side. What remains is the equation

$$1 = \int dx \delta(x - x'). \quad (2.42)$$

Thus there is unit area under the graph of $\delta(x)$, which is remarkable, given that the function vanishes for $x \neq 0$! Although the name of $\delta(x)$ includes the word 'function', this object is not really a function because we cannot assign it a value at the origin. It is best considered to be the limit of a

⁴ The analogy would be clearer if we wrote $a(x)$ for $\psi(x)$, but for historical reasons the letter ψ is hard to avoid in this context.

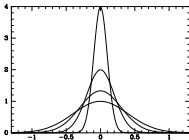


Figure 2.1 A series of Gaussians of unit area. The Dirac delta function is the limit of this series of functions as the dispersion tends to zero.

series of functions that all have unit area under their graphs but become more and more sharply peaked around the origin (see Figure 2.1).

The analogue of equation (1.27) is

$$\int dx |\psi(x)|^2 = 1, \quad (2.43)$$

which expresses the physical requirement that there is unit probability of finding the particle at *some* value of x .

The analogue of equation (2.2) is

$$I = \int dx |x\rangle\langle x|. \quad (2.44)$$

It is instructive to check that the operator that is defined by the right side of this equation really is the identity operator. Applying the operator to an arbitrary state $|\psi\rangle$ we find

$$I|\psi\rangle = \int dx |x\rangle\langle x|\psi\rangle. \quad (2.45)$$

By equations (2.37) and (2.38) the expression on the right of this equation is $|\psi\rangle$, so I is indeed the identity operator.

When we multiply (2.45) by $\langle\phi|$ on the left, we obtain an important formula

$$\langle\phi|\psi\rangle = \int dx \langle\phi|x\rangle\langle x|\psi\rangle = \int dx \phi^*(x)\psi(x), \quad (2.46)$$

where the second equality uses equations (2.38) and (2.39). Many practical problems reduce to the evaluation of an amplitude such as $\langle\phi|\psi\rangle$. The expression on the right of equation (2.46) is a well-defined integral that evaluates to the desired number.

Box 2.2: Proof that $\hat{p}$ is Hermitian

We have to show that for any states $|\phi\rangle$ and $|\psi\rangle$, $\langle\psi|\hat{p}|\phi\rangle = (\langle\phi|\hat{p}|\psi\rangle)^*$. We use equation (2.49) to write the left side of this equation in the position representation:

$$\langle\psi|\hat{p}|\phi\rangle = -i\hbar \int dx \psi^*(x) \frac{\partial\phi}{\partial x}.$$

Integrating by parts this becomes

$$\langle\psi|\hat{p}|\phi\rangle = -i\hbar \left([\psi^*\phi]_{-\infty}^{\infty} - \int dx \phi(x) \frac{\partial\psi^*}{\partial x} \right).$$

We assume that all wavefunctions vanish at spatial infinity, so the term in square brackets vanishes, and

$$\langle\psi|\hat{p}|\phi\rangle = i\hbar \int dx \phi(x) \frac{\partial\psi^*}{\partial x} = (\langle\phi|\hat{p}|\psi\rangle)^*.$$

By analogy with equation (2.5), the **position operator** is

$$\hat{x} = \int dx x|x\rangle\langle x|. \quad (2.47)$$

After applying $\hat{x}$ to a ket $|\psi\rangle$ we have a ket $|\phi\rangle = \hat{x}|\psi\rangle$ whose wavefunction $\phi(x') = \langle x'|\hat{x}|\psi\rangle$ is

$$\begin{aligned} \phi(x') &= \langle x'|\hat{x}|\psi\rangle = \int dx x \langle x'|x\rangle\langle x|\psi\rangle \\ &= \int dx x \delta(x - x')\psi(x) = x'\psi(x'), \end{aligned} \quad (2.48)$$

where we have used equations (2.38) and (2.40). Equation (2.48) states that the operator $\hat{x}$ simply multiplies the wavefunction $\psi(x)$ by its argument.

In the position representation, operators turn functions of x into other functions of x . An easy way of making a new function out of an old one is to differentiate it. So consider the operator $\hat{p}$ that is defined by

$$\langle x|\hat{p}|\psi\rangle = (\hat{p}\psi)(x) = -i\hbar \frac{\partial\psi}{\partial x}. \quad (2.49)$$

In Box 2.2 we show that the factor i ensures that $\hat{p}$ is a Hermitian operator. The factor $\hbar$ ensures that $\hat{p}$ has the dimensions of momentum:⁵ we will find that $\hat{p}$ is the **momentum operator**. In Newtonian physics the momentum of a particle of mass m and velocity $\dot{x}$ is $m\dot{x}$, so let's use equation (2.34) to calculate $d\langle\hat{x}\rangle/dt$ and see whether it is $\langle\hat{p}\rangle/m$.

⁵ Planck's constant $\hbar = 2\pi\hbar$ has dimensions of distance $\times$ momentum, or, equivalently, energy $\times$ time, or, most simply, angular momentum.

2.3.1 Hamiltonian of a particle

To calculate any time derivatives in quantum mechanics we need to know what the Hamiltonian operator H of our system is because H appears in the TDSE (2.26). Equation (2.5) defines H in the energy representation, but not how to write H in the position representation. We are going to have to make an informed guess and justify our guess later.

The Newtonian expression for the energy of a particle is

$$E = \frac{1}{2}mv^2 + V = \frac{p^2}{2m} + V, \quad (2.50)$$

where $V(x)$ is the particle's potential energy. So we guess that the Hamiltonian of a particle is

$$H = \frac{\hat{p}^2}{2m} + V(\hat{x}), \quad (2.51)$$

where the square of $\hat{p}$ means the act of operating with $\hat{p}$ twice ($\hat{p}^2 = \hat{p}\hat{p}$). The meaning of $V(\hat{x})$ is given by equation (2.20) with V and $\hat{x}$ substituted for f and R . Working from that equation in close analogy with the calculation in equation (2.48) demonstrates that in the position representation the operator $V(\hat{x})$ acts on a wavefunction $\psi(x)$ simply by multiplying ψ by $V(x)$. That is, $\langle x|V(\hat{x})|\psi\rangle = V(x)\psi(x)$.

Now that we have guessed that H is given by equation (2.51), the next step in the calculation of the rate of change of $\langle\hat{x}\rangle$ is to evaluate the commutator of $\hat{x}$ and H . Making use of equations (2.22) we find

$$\begin{aligned} [\hat{x}, H] &= \left[\hat{x}, \frac{\hat{p}^2}{2m} + V \right] = \frac{[\hat{x}, \hat{p}\hat{p}]}{2m} + [\hat{x}, V(\hat{x})] \\ &= \frac{[\hat{x}, \hat{p}]\hat{p} + \hat{p}[\hat{x}, \hat{p}]}{2m}. \end{aligned} \quad (2.52)$$

In the last equality we have used the fact that $[\hat{x}, V(\hat{x})] = 0$, which follows because both $\hat{x}$ and $V(\hat{x})$ act by multiplication, and ordinary multiplication is a commutative operation. We now have to determine the value of the commutator $[\hat{x}, \hat{p}]$. We return to the definition (2.49) of $\hat{p}$ and calculate the wavefunction produced by applying $[\hat{x}, \hat{p}]$ to an arbitrary state $|\psi\rangle$

$$\begin{aligned} \langle x|[\hat{x}, \hat{p}]|\psi\rangle &= \langle x|(\hat{x}\hat{p} - \hat{p}\hat{x})|\psi\rangle = -i\hbar \left(x \frac{\partial\psi}{\partial x} - \frac{\partial(x\psi)}{\partial x} \right) \\ &= i\hbar \langle x|\psi\rangle. \end{aligned} \quad (2.53)$$

Since this equation holds for any $|\psi\rangle$, we have the operator equation

$$[\hat{x}, \hat{p}] = i\hbar. \quad (2.54)$$

This key result, that the commutator of $\hat{x}$ with $\hat{p}$ is the constant $i\hbar$, is called a **canonical commutation relation**.⁶ Two observables whose

⁶ The name derives from 'canonical coordinates' in Hamilton's formulation of classical mechanics.

commutator is $\pm i\hbar$ are said to be **canonically conjugate** to one another, or **conjugate observables**.

Finally we have the hoped-for relation between p and $\dot{x}$: substituting equations (2.53) and (2.54) into equation (2.34) we have

$$\begin{aligned}\frac{d\langle x \rangle}{dt} &= \frac{d}{dt} \langle \psi | \hat{x} | \psi \rangle = -\frac{i}{\hbar} \langle \psi | [\hat{x}, H] | \psi \rangle = -\frac{i}{\hbar} \frac{i\hbar}{m} \langle \psi | \hat{p} | \psi \rangle \\ &= \frac{1}{m} \langle \hat{p} \rangle.\end{aligned}\quad (2.55)$$

This result makes it highly plausible that $\hat{p}$ is indeed the momentum operator.

A calculation of the rate of change of $\langle \hat{p} \rangle$ will increase the plausibility still further. Again working from (2.34) and using (2.51) we have

$$\frac{d\langle \hat{p} \rangle}{dt} = -\frac{i}{\hbar} \langle [\hat{p}, H] \rangle = -\frac{i}{\hbar} \langle [\hat{p}, V] \rangle. \quad (2.56)$$

Since $[\hat{p}, \hat{x}] = -i\hbar$ is just a number, equation (2.25) for the commutator of one operator with a function of another operator can be used to evaluate $[\hat{p}, V(\hat{x})]$. We then have

$$\frac{d\langle \hat{p} \rangle}{dt} = -\left\langle \frac{dV}{d\hat{x}} \right\rangle. \quad (2.57)$$

That is, the expectation of the rate of change of the momentum is equal to the expectation of the force on the particle. Thus we have recovered Newton's second law from the TDSE. This achievement gives us confidence that (2.51) is the correct expression for H .

2.3.2 Wavefunction for well-defined momentum

From the discussion below equation (2.11) we know that the state $|p\rangle$ in which a measurement of the momentum will certainly yield the value p has to be an eigenstate of $\hat{p}$. We find the wavefunction $u_p(x) = \langle x | p \rangle$ of this important state by using equation (2.49) to write the defining equation $\hat{p}|p\rangle = p|p\rangle$ in the position representation:

$$\langle x | \hat{p} | p \rangle = -i\hbar \frac{\partial u_p}{\partial x} = p \langle x | p \rangle = p u_p(x). \quad (2.58)$$

The solution of this differential equation is

$$u_p(x) = A e^{ipx/\hbar}. \quad (2.59)$$

Hence the wavefunction of a particle of well-defined momentum is a plane wave with wavelength $\lambda = 2\pi/k = h/\sqrt{2mE}$, where m is the particle's

mass and E its kinetic energy; λ is called the particle's **de Broglie wavelength**.⁷

If we try to choose the constant A in (2.59) to ensure that u_p satisfies the usual normalisation condition (2.43), we will fail because the integral over all x of $|e^{ipx/\hbar}|^2 = 1$ is undefined. Instead we choose A as follows. By analogy with (2.40) we require $\langle p'|p \rangle = \delta(p - p')$. When we use (2.44) to insert an identity operator into this expression, it becomes

$$\delta(p - p') = \int dx \langle p'|x \rangle \langle x|p \rangle = |A|^2 \int dx e^{i(p-p')x/\hbar} = 2\pi\hbar |A|^2 \delta(p - p'), \quad (2.60)$$

where we have used equation (C.12) to evaluate the integral. Thus $|A|^2 = \hbar^{-1}$, where $\hbar = 2\pi\hbar$ is Planck's constant, and the correctly normalised wavefunction of a particle of momentum p is

$$u_p(x) = \langle x|p \rangle = \frac{1}{\sqrt{\hbar}} e^{ipx/\hbar}. \quad (2.61)$$

The uncertainty principle It follows from (2.61) that the position of a particle that has well-defined momentum is maximally uncertain: all values of x are equally probable because $|u_p|^2$ is independent of x . This phenomenon is said to be a consequence of the **uncertainty principle**,⁸ namely that when an observable has a well-defined value, all values of the canonically conjugate observable are equally probable.

The complex conjugate of equation (2.61) is

$$\langle p|x \rangle = \frac{1}{\sqrt{\hbar}} e^{-ipx/\hbar}, \quad (2.62)$$

which states that when the particle's position is certain to be x , the amplitude to measure momentum p is proportional to $e^{-ipx/\hbar}$. It follows that when the position is certain, all momenta are equally probable.

From equations (2.61) and (2.62) it follows that states in which either a particle's position or its momentum is certain are unphysical, for they say that a measurement that put the particle into the state $|x\rangle$ would have to knock the particle so hard that after the measurement its momentum might be relativistic (cf. the discussion of Heisenberg's microscope on page 7), while a measurement that put the particle into the state $|p\rangle$ might scatter the particle into the furthest corner of the universe.

So let us now investigate realistic states, namely ones in which the results of measurements of both position and momentum are uncertain.

⁷ Louis de Broglie (1892–1987) was the second son of the Duc de Broglie. In 1924 his PhD thesis introduced the concept of matter waves, by considering relativistic invariance of phase. For this work he won the 1929 Nobel Prize for physics. In later years he struggled to find a causal rather than probabilistic interpretation of quantum mechanics.

⁸ First stated by Werner Heisenberg, *Z. Phys.*, **43**, 172 (1927), and consequently often called 'Heisenberg's uncertainty principle'.

Box 2.3: Gaussian integrals

Consider the integral

$$I = \int_{-\infty}^{\infty} dx e^{-(b^2 x^2 + ax)}, \quad (1)$$

where a and b are constants. We observe that $b^2 x^2 + ax = (bx + a/2b)^2 - a^2/4b^2$. Thus we may write $I = e^{a^2/4b^2} b^{-1} \int dz e^{-z^2}$, where $z = bx + a/2b$. The integral is equal to $\sqrt{\pi}$. Hence we have the very useful result

$$\int_{-\infty}^{\infty} dx e^{-(b^2 x^2 + ax)} = \frac{\sqrt{\pi}}{b} e^{a^2/4b^2}. \quad (2)$$

For definiteness we take the probability density $|\psi(x)|^2$ to be a Gaussian distribution of dispersion σ . So we write

$$\psi(x) = \frac{1}{(2\pi\sigma^2)^{1/4}} e^{-x^2/4\sigma^2}. \quad (2.63)$$

With equations (2.46) and (2.61) we find that in this state the amplitude to measure momentum p is

$$\langle p|\psi\rangle = \int dx u_p^*(x)\psi(x) = \frac{1}{\sqrt{h}(2\pi\sigma^2)^{1/4}} \int dx e^{-ipx/h} e^{-x^2/4\sigma^2}. \quad (2.64)$$

Box 2.3 explains how integrals of this type are evaluated. Setting $a = ip/h$ and $b = (2\sigma)^{-1}$ in equation (2) of Box 2.3 we find

$$\langle p|\psi\rangle = \frac{2\sigma\sqrt{\pi}}{\sqrt{h}(2\pi\sigma^2)^{1/4}} e^{-\sigma^2 p^2/h^2} = \frac{1}{(2\pi h^2/4\sigma^2)^{1/4}} e^{-\sigma^2 p^2/h^2}. \quad (2.65)$$

The probability density $|\langle p|\psi\rangle|^2$ is a Gaussian centred on zero with a dispersion σ_p in p that equals $\hbar/2\sigma$. Thus, the more sharply peaked the particle's probability distribution is in x , the broader the distribution is in p . The product of the dispersions in x and p is always $\frac{1}{2}\hbar$: $\sigma_p\sigma = \frac{1}{2}\hbar$.

This trade-off between the uncertainties in x and p arises because when we expand $|\psi\rangle$ in eigenkets of $\hat{p}$, localisation of the probability amplitude $\psi(x)$ is caused by interference between states of different momenta: in the position representation, these states are plane waves of wavelength $\hbar/p$ that have the same amplitude everywhere, and interference between waves of very different wavelengths is required if the region of constructive interference is to be strongly confined.

2.3.3 Dynamics of a free particle

We now consider the motion of a free particle – one that is subject to no forces so we can drop the potential term in the Hamiltonian (2.51). Consequently, the Hamiltonian of a free particle,

$$H = \frac{\hat{p}^2}{2m}, \quad (2.66)$$

is a function of $\hat{p}$ alone, so its eigenkets will be the eigenkets (2.61) of $\hat{p}$. By expressing any ket $|\psi\rangle$ as a linear combination of these eigenkets, and using the basic time-evolution equation (2.32), we can follow the motion of a particle from the initial state $|\psi\rangle$. We illustrate this procedure with the case in which ψ corresponds to the particle being approximately at the origin with momentum near some value p_0 . Equation (2.65) gives $\langle p|\psi\rangle$ for the case in which p_0 vanishes. The amplitude distribution that we require is

$$\langle p|\psi, 0\rangle = \frac{1}{(2\pi\hbar^2/4\sigma^2)^{1/4}} e^{-\sigma^2(p-p_0)^2/\hbar^2}. \quad (2.67)$$

We can now use (2.32) to obtain the wavefunction t units of time later

$$\begin{aligned} \langle x|\psi, t\rangle &= \int dp \langle x|p\rangle \langle p|\psi, 0\rangle e^{-ip^2t/2m\hbar} \\ &= \frac{1}{\sqrt{\hbar}(2\pi\hbar^2/4\sigma^2)^{1/4}} \int dp e^{ipx/\hbar} e^{-\sigma^2(p-p_0)^2/\hbar^2} e^{-ip^2t/2m\hbar}. \end{aligned} \quad (2.68)$$

Evaluating the integral in this expression involves some tiresome algebra – you can find the details in Box 2.4 if you are interested. We want the probability density at time t of finding the particle at x , which is the mod-square of equation (2.68). From the last equation of Box 2.4 we have

$$|\langle x|\psi, t\rangle|^2 = \frac{\sigma}{\sqrt{2\pi\hbar^2}|b|^2} \exp\left\{-\frac{(x-p_0t/m)^2\sigma^2}{2\hbar^4|b|^4}\right\}. \quad (2.69)$$

This is a Gaussian distribution whose centre moves with the velocity p_0/m associated with the most probable momentum in the initial data (2.67). The square of the Gaussian's dispersion is

$$\sigma^2(t) = \sigma^2 + \left(\frac{\hbar t}{2m\sigma}\right)^2. \quad (2.70)$$

We saw above that in the initial data the uncertainty in p is $\sim \sigma_p = \hbar/2\sigma$, which translates to an uncertainty in velocity $\Delta_v \sim \hbar/2m\sigma$. After time t this uncertainty should lead to an additional uncertainty in position $\Delta_x = \Delta_v t \sim \hbar t/2m\sigma$ in perfect agreement with equation (2.70).

These results complete the demonstration that the identification of the operator $\hat{p}$ defined by equation (2.49) with the momentum operator,

Box 2.4: Evaluating the integral in equation (2.68)

The integral is of the form discussed in Box 2.3. To clean it up we replace the p^2 in the third exponential with $(p - p_0)^2 + 2p_0p - p_0^2$ and gather together all three exponents:

$$\begin{aligned} \langle x|\psi, t\rangle &= \frac{e^{ip_0t/2m\hbar}}{\sqrt{\hbar(2\pi\hbar^2/4\sigma^2)^{1/4}}} \\ &\times \int dp \exp\left\{\frac{ip}{\hbar}\left(x - \frac{p_0t}{m}\right) - (p - p_0)^2\left(\frac{\sigma^2}{\hbar^2} + \frac{it}{2m\hbar}\right)\right\}. \end{aligned}$$

In Box 2.3 we now set

$$a = \frac{i}{\hbar}\left(x - \frac{p_0t}{m}\right) \quad ; \quad b^2 = \left(\frac{\sigma^2}{\hbar^2} + \frac{it}{2m\hbar}\right)$$

and conclude that

$$\langle x|\psi, t\rangle = \frac{e^{ip_0t/2m\hbar}}{\sqrt{\hbar(2\pi\hbar^2/4\sigma^2)^{1/4}}} \exp\left\{\frac{ip_0}{\hbar}\left(x - \frac{p_0t}{m}\right)\right\} \frac{\sqrt{\pi}}{b} e^{-(x - p_0t/m)^2/4\hbar^2b^2}.$$

This is a complicated result because b is a complex number, but its mod-square, equation (2.69), is relatively simple.

together with the Hamiltonian (2.51), enable us to recover as much of Newtonian mechanics as we expect to continue valid outside the classical regime. The idea that in an appropriate limit the predictions of quantum mechanics should agree with classical mechanics is often called the **correspondence principle**. The discipline of checking that one's calculations comply with the correspondence principle is useful in several ways: (i) it provides a check on the calculations, helping one to locate missing factors of i or incorrect signs, (ii) it deepens one's understanding of classical physics, and (iii) it draws attention to novel predictions of quantum mechanics that have no counterparts in classical mechanics.

In the process of checking the correspondence principle for a free particle we have stumbled on a new principle, the uncertainty principle, which implies that the more tightly constrained the value of one observable is, the more uncertain the value of the conjugate variable must be. Notice that these uncertainties do not arise from measurement errors: we have assumed that x and p can be measured exactly. The uncertainties we have discussed are inherent in the situation and can only be increased by deficiencies in the measurement process.

Our calculations have also shown how far-reaching the principle of quantum interference is: equation (2.68), upon which our understanding of the dynamics of a free particle rests, expresses the amplitude for the particle to be found at x at time t as an integral over momenta of the amplitude to travel at momentum p . It is through interference between the infinite number of contributing amplitudes that classically recognisable dynamics is recovered. Had we mod-squared the amplitudes before

adding them, as classical probability theory would suggest, we would have obtained entirely unphysical results.

2.3.4 Back to two-slit interference

When we discussed the two-slit interference experiment in §1.3.1, we stated without proof that $\phi_1 - \phi_2 \propto x$, where $\phi_i(x)$ is the phase of the amplitude $A_i(x)$ for an electron to arrive at the point x on the screen P after passing through the slit S_i . We can now justify this assertion and derive the constant of proportionality. Once the constant has been determined, it is possible to assess the feasibility of the experiment from a practical point of view.

We assume that the quantum state of an electron as it emerges from the electron gun can be approximated by a state of well-defined momentum $|p\rangle$. So the wavefunction between the gun and the screen with the slits is a plane wave of wavelength $\lambda = h/p$. As an electron passes through a slit we assume that it is deflected slightly but retains its former kinetic energy. So we approximate its wavefunction after passing through the slit by a wave that is no longer plane, but still has wavelength λ . Hence the phase of this wave at position x on the screen P will be the phase at the slit plus $2\pi D/\lambda = pD/h$, where $D(x)$ is the distance from x to the slit. By Pythagoras's theorem

$$D = \sqrt{L^2 + (x \pm s)^2}, \quad (2.71)$$

where L is the distance between the screen with the slits and P, $2s$ is the distance between the slits, and the plus sign applies for one slit and the minus sign to the other. We assume that both x and s are much smaller than L so the square root can be expanded by the binomial theorem. We then find that the difference of the phases is

$$\phi_1 - \phi_2 \simeq \frac{2psx}{hL}. \quad (2.72)$$

The distance X between the dark bands on P is the value of x for which the left side becomes 2π , so

$$X = \frac{hL}{2ps}. \quad (2.73)$$

Let's put some numbers into this formula. Since $h = 6.63 \times 10^{-34}$ J s is very small, there is a danger that X will come out too small to produce observable bands. Therefore we choose L fairly large and both p and s small. Suppose we adopt 1 m for L and 1 μ m for s . From the Hamiltonian (2.66) we have $p = \sqrt{2mE}$. A reasonable energy for the electrons is $E = 100$ eV = 1.6×10^{-17} J, which yields $p = 5.5 \times 10^{-24}$ N s, and $X = 0.057$ mm. Hence there should be no difficulty observing a sinusoidal pattern that has this period.

What do the numbers look like for bullets? On a firing range we can probably stretch L to 1000 m. The distance between the slits clearly has

to be larger than the diameter of a bullet, so we take $s = 1$ cm. A bullet weighs ~ 10 gm and travels at ~ 300 m s $^{-1}$. Equation (2.73) now yields $X \sim 10^{-29}$ m. So it is not surprising that fire-arms manufacturers find classical mechanics entirely satisfactory.

2.3.5 Generalisation to three dimensions

Real particles move in three dimensions rather than one. Fortunately, the generalisation to three dimensions of what we have done in one dimension is straightforward.

The x , y and z coordinates of a particle are three distinct observables. Their operators commute with one another:

$$[\hat{x}_i, \hat{x}_j] = 0. \quad (2.74)$$

Since these are commuting observables, there is a complete set of mutual eigenkets, $\{|\mathbf{x}\rangle\}$. We can express any state of the system, $|\psi\rangle$, as a linear combination of these kets:

$$|\psi\rangle = \int d^3\mathbf{x} \langle\mathbf{x}|\psi\rangle |\mathbf{x}\rangle = \int d^3\mathbf{x} \psi(\mathbf{x}) |\mathbf{x}\rangle, \quad (2.75)$$

where the wavefunction $\psi(\mathbf{x})$ is now a function of three variables, and the integral is over all space.

The x , y and z coordinates of the particle's momentum $\hat{\mathbf{p}}$ commute with one another:

$$[\hat{p}_i, \hat{p}_j] = 0. \quad (2.76)$$

In the position representation, these operators are represented by partial derivatives with respect to their respective coordinates

$$\hat{p}_i = -i\hbar \frac{\partial}{\partial x_i} \quad \text{so} \quad \hat{\mathbf{p}} = -i\hbar \nabla. \quad (2.77)$$

The momenta commute with all operators except their conjugate coordinate, so the canonical commutation relations are

$$[\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij}. \quad (2.78)$$

In §4.2 we will understand the origin of the factor δ_{ij} . Since the three momentum operators commute with one another, there is a complete set of mutual eigenstates. Analogously to equation (2.61), the wavefunction of the state with well-defined momentum $\mathbf{p}$ is

$$\langle\mathbf{x}|\mathbf{p}\rangle = \frac{1}{h^{3/2}} e^{i\mathbf{x}\cdot\mathbf{p}/\hbar}. \quad (2.79)$$

In the position representation the TDSE of a particle of mass m that moves in a potential $V(\mathbf{x})$ reads

$$i\hbar \frac{\partial \langle\mathbf{x}|\psi\rangle}{\partial t} = \langle\mathbf{x}|\hat{H}|\psi\rangle = \langle\mathbf{x}|\frac{\hat{p}^2}{2m}|\psi\rangle + \langle\mathbf{x}|V(\hat{\mathbf{x}})|\psi\rangle. \quad (2.80)$$

Now $\langle \mathbf{x} | \hat{p}^2 | \psi \rangle = -\hbar^2 \nabla^2 \langle \mathbf{x} | \psi \rangle$ and $\langle \mathbf{x} | V(\hat{\mathbf{x}}) | \psi \rangle = V(\mathbf{x}) \langle \mathbf{x} | \psi \rangle$. Hence using the definition $\psi(\mathbf{x}) = \langle \mathbf{x} | \psi \rangle$, the TDSE becomes

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(\mathbf{x})\psi. \quad (2.81)$$

Probability current Max Born⁹ first suggested that the mod-square of a particle's wavefunction,

$$\rho(x, t) = |\psi(\mathbf{x}, t)|^2 \quad (2.82)$$

is the probability density of finding the particle near $\mathbf{x}$ at time t . Since the particle is certain to be found *somewhere*, this interpretation implies that at any time $\int d^3\mathbf{x} |\psi(\mathbf{x}, t)|^2 = 1$. It is not self-evident that this physical requirement is satisfied when the wavefunction evolves according to the TDSE (2.81). We now show that it is in fact satisfied.

We multiply the TDSE (2.81) by ψ^* and subtract from it the result of multiplying the complex conjugate of (2.81) by ψ . Then the terms involving the potential $V(\mathbf{x})$ cancel and we are left with

$$i\hbar \left(\psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t} \right) = -\frac{\hbar^2}{2m} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*). \quad (2.83)$$

The left side of this equation is a multiple of the time derivative of $\rho = \psi^* \psi$. The right side can be expressed as a multiple of the divergence of the **probability current**

$$\mathbf{J}(\mathbf{x}) = \frac{i\hbar}{2m} (\psi \nabla \psi^* - \psi^* \nabla \psi). \quad (2.84)$$

That is, equation (2.83) can be written

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}. \quad (2.85)$$

In fluid mechanics this equation with $J = \rho \mathbf{v}$ expresses the conservation of mass as a fluid of density ρ flows with velocity $\mathbf{v}(\mathbf{x})$. In quantum mechanics it expresses conservation of probability. To show that this is so, we simply integrate both sides of equation (2.85) through a volume $\mathcal{V}$. Then we obtain

$$\frac{d}{dt} \int_{\mathcal{V}} d^3\mathbf{x} \rho = \int_{\mathcal{V}} d^3\mathbf{x} \frac{\partial \rho}{\partial t} = - \int_{\mathcal{V}} d^3\mathbf{x} \nabla \cdot \mathbf{J} = - \oint_{\partial\mathcal{V}} d^2\mathbf{S} \cdot \mathbf{J}, \quad (2.86)$$

where the last equality uses the divergence theorem and $\partial\mathcal{V}$ denotes the boundary of $\mathcal{V}$. Equation (2.86) states that the rate of increase of the

⁹ For this insight Born won the 1954 Nobel Prize for physics. In fact the text of the key paper (M. Born, *Z. Physik*, **37** 863 (1926)) argues that ψ is the probability density, but a note in proof says "On more careful consideration, the probability is proportional to the square of ψ ".

probability $P = \int_V d^3\mathbf{x} \rho$ of finding the particle in V is equal to minus the integral over the volume's bounding surface of the probability flux out of the volume. If V encompasses all space, ψ , and therefore $\mathbf{J}$, will vanish on the boundary, so $\int_V d^3\mathbf{x} \rho$ will be constant.

We can gain valuable insight into the meaning of a wavefunction by explicitly breaking ψ into its modulus and phase:

$$\psi(\mathbf{x}) = S(\mathbf{x})e^{i\phi(\mathbf{x})}, \quad (2.87)$$

where S and ϕ are real. Substituting this expression into the definition (2.84) of $\mathbf{J}$, we find

$$\mathbf{J} = \frac{i\hbar}{2m}(S\nabla S - iS^2\nabla\phi - S\nabla S - iS^2\nabla\phi) = \frac{\hbar}{m}S^2\nabla\phi. \quad (2.88)$$

Since $S^2 = |\psi|^2 = \rho$, the velocity $\mathbf{v}$ that is defined by setting $\mathbf{J} = \rho\mathbf{v}$ is

$$\mathbf{v} = \frac{\hbar\nabla\phi}{m}. \quad (2.89)$$

Thus the gradient of the phase of the wavefunction encodes the velocity at which the probability fluid flows. In classical physics, this is the particle's velocity. The phase of the wavefunction (2.79) of a particle of well-defined momentum is $\phi(\mathbf{x}) = \mathbf{x}\cdot\mathbf{p}/\hbar$, so in this special case $\mathbf{v} = \mathbf{p}/m$ as in classical physics. Equation (2.89) extends the connection between velocity and the gradient of phase to general wavefunctions.

The virial theorem We illustrate the use of the canonical commutations relations (2.74), (2.76) and (2.78) by deriving a relation between the kinetic and potential energies of a particle that is in a stationary state. In §2.2.1 we showed that all expectation values are time-independent when a system is in a stationary state. We apply this result to the operator $\hat{\mathbf{x}} \cdot \hat{\mathbf{p}}$

$$\begin{aligned} 0 &= i\hbar \frac{d}{dt} \langle \hat{\mathbf{x}} \cdot \hat{\mathbf{p}} \rangle = \langle E | \left[\hat{\mathbf{x}} \cdot \hat{\mathbf{p}}, \frac{\hat{p}^2}{2m} + V(\hat{\mathbf{x}}) \right] | E \rangle \\ &= \frac{1}{2m} \langle E | [\hat{\mathbf{x}} \cdot \hat{\mathbf{p}}, \hat{p}^2] | E \rangle + \langle E | [\hat{\mathbf{x}} \cdot \hat{\mathbf{p}}, V(\hat{\mathbf{x}})] | E \rangle. \end{aligned} \quad (2.90)$$

The first commutator can be expanded thus

$$[\hat{\mathbf{x}} \cdot \hat{\mathbf{p}}, \hat{p}^2] = \sum_{jk} [\hat{x}_j \hat{p}_j, \hat{p}_k^2] = \sum_{jk} [\hat{x}_j, \hat{p}_k^2] \hat{p}_j = \sum_{jk} 2i\hbar \hat{p}_k \delta_{jk} \hat{p}_j = 2i\hbar \hat{p}^2. \quad (2.91)$$

In the position representation the second commutator is simply

$$[\hat{\mathbf{x}} \cdot \hat{\mathbf{p}}, V(\hat{\mathbf{x}})] = -i\hbar \mathbf{x} \cdot \nabla V(\mathbf{x}). \quad (2.92)$$

When we put these results back into (2.90) and rearrange, we obtain the **virial theorem**

$$2\langle E | \frac{\hat{p}^2}{2m} | E \rangle = \langle E | (\mathbf{x} \cdot \nabla V) | E \rangle. \quad (2.93)$$

In important applications the potential is proportional to some power of distance from the origin: $V(\mathbf{x}) = C|\mathbf{x}|^\alpha$. Then, because $\nabla|\mathbf{x}| = \mathbf{x}/|\mathbf{x}|$, the operator on the right is $\mathbf{x} \cdot \nabla V = \alpha C|\mathbf{x}|^\alpha = \alpha V$ and the virial theorem becomes

$$2\langle E | \frac{\hat{p}^2}{2m} | E \rangle = \alpha \langle E | V | E \rangle. \quad (2.94)$$

So twice the kinetic energy is equal to α times the potential energy. For example, for a harmonic oscillator $\alpha = 2$, so the kinetic and potential energies are equal. The other important example is motion in an inverse-square force field, such as the electrostatic field of an atomic nucleus. In this case $\alpha = -1$, so twice the kinetic energy plus the potential energy vanishes. Equivalently, the kinetic energy is equal in magnitude but opposite in sign to the total energy.

2.4 Summary

Every observable is associated with an operator. The eigenkets of this operator are the states of the system in which the result of measuring the observable are certain, and its eigenvalues are the numerical values that may be measured. If these values are real numbers, the operator is Hermitian.

The Hamiltonian is the operator that's associated with energy, and it has a special status because it determines how a state evolves in time. We predict the future of a system by solving the TDSE for its state $|\psi, t\rangle$. The temporal evolution of states of well-defined energy – stationary states – is trivial, and by expressing any given initial state as a linear combination of stationary states, we can easily determine the system's future states. Consequently, much of quantum mechanics focuses on identifying stationary states. These states owe their name to the fact that in such a state the expectation value of any observable is time-independent.

We recovered classical mechanics from quantum mechanics in the form of Ehrenfest's theorem.

In the position representation we use as a complete set of amplitudes for a particle the amplitudes $\psi(x) = \langle x | \psi \rangle$ to be found at x . To specify the state of a particle in classical mechanics we have to specify its momentum p in addition to its position, but in quantum mechanics the state is specified by an entire function $\langle x | \psi \rangle$ rather than two numbers, and from this function we can compute the amplitudes $\langle p | \psi \rangle$ for measuring any momentum. Since states of well-defined momentum are sinusoidal waves, if the momentum is well defined, the position is totally uncertain, and vice versa. Gaussian wavefunctions $\langle x | \psi \rangle$ yield Gaussian momentum distributions $\langle p | \psi \rangle$ and the dispersions of the two resulting probability distributions satisfy the uncertainty relation $\sigma_x \sigma_p = \hbar/2$. When a particle moves freely, the uncertainty in its position grows steadily with time because the uncertainty in its momentum gives rise to an ever-increasing uncertainty in the distance it has travelled.

The motion of a particle can be most completely described by the probability current $\mathbf{J}$, which is $\hbar/m$ times the probability density $|\psi|^2$ times the gradient of the wavefunction's phase.

Problems

2.1 How is a wavefunction $\psi(x)$ written in Dirac's notation? What's the physical significance of the complex number $\psi(x)$ for given x ?

2.2 Let Q be an operator. Under what circumstances is the complex number $\langle a|Q|b\rangle$ equal to the complex number $(\langle b|Q|a\rangle)^*$ for any states $|a\rangle$ and $|b\rangle$?

2.3 Let Q be the operator of an observable and let $|\psi\rangle$ be the state of our system.

- What are the physical interpretations of $\langle\psi|Q|\psi\rangle$ and $|\langle q_n|\psi\rangle|^2$, where $|q_n\rangle$ is the n^{th} eigenket of the observable Q and q_n is the corresponding eigenvalue?
- What is the operator $\sum_n |q_n\rangle\langle q_n|$, where the sum is over all eigenkets of Q ? What is the operator $\sum_n q_n |q_n\rangle\langle q_n|$?
- If $u_n(x)$ is the wavefunction of the state $|q_n\rangle$, write down an integral that evaluates to $\langle q_n|\psi\rangle$.

2.4 What does it mean to say that two operators commute? What is the significance of two observables having mutually commuting operators?

Given that the commutator $[P, Q] \neq 0$ for some observables P and Q , does it follow that for all $|\psi\rangle \neq 0$ we have $[P, Q]|\psi\rangle \neq 0$?

2.5 Let $\psi(x, t)$ be the correctly normalised wavefunction of a particle of mass m and potential energy $V(x)$. Write down expressions for the expectation values of (a) x ; (b) x^2 ; (c) the momentum p_x ; (d) p_x^2 ; (e) the energy.

What is the probability that the particle will be found in the interval (x_1, x_2) ?

2.6 Write down the time-independent (TISE) and the time-dependent (TDSE) Schrödinger equations. Is it necessary for the wavefunction of a system to satisfy the TDSE? Under what circumstances does the wavefunction of a system satisfy the TISE?

2.7 Why is the TDSE first-order in time, rather than second-order like Newton's equations of motion?

2.8 A particle is confined in a potential well such that its allowed energies are $E_n = n^2\mathcal{E}$, where $n = 1, 2, \dots$ is an integer and $\mathcal{E}$ a positive constant. The corresponding energy eigenstates are $|1\rangle, |2\rangle, \dots, |n\rangle, \dots$. At $t = 0$ the particle is in the state

$$|\psi(0)\rangle = 0.2|1\rangle + 0.3|2\rangle + 0.4|3\rangle + 0.843|4\rangle. \quad (2.95)$$

- What is the probability, if the energy is measured at $t = 0$, of finding a number smaller than $6\mathcal{E}$?
- What is the mean value and what is the rms deviation of the energy of the particle in the state $|\psi(0)\rangle$?
- Calculate the state vector $|\psi\rangle$ at time t . Do the results found in (a) and (b) for time t remain valid for arbitrary time t ?
- When the energy is measured it turns out to be $16\mathcal{E}$. After the measurement, what is the state of the system? What result is obtained if the energy is measured again?

2.9 A system has a time-independent Hamiltonian that has spectrum $\{E_n\}$. Prove that the probability P_k that a measurement of energy will yield the value E_k is time-independent. Hint: you can do this either from Ehrenfest's theorem, or by differentiating $\langle E_k, t | \psi \rangle$ w.r.t. t and using the TDSE.

2.10 Let $\psi(x)$ be a properly normalised wavefunction and Q an operator on wavefunctions. Let $\{q_r\}$ be the spectrum of Q and $\{u_r(x)\}$ be the corresponding correctly normalised eigenfunctions. Write down an expression for the probability that a measurement of Q will yield the value q_r . Show that $\sum_r P(q_r | \psi) = 1$. Show further that the expectation of Q is $\langle Q \rangle = \int_{-\infty}^{\infty} \psi^* Q \psi dx$.¹⁰

2.11 Find the energy of neutron, electron and electromagnetic waves of wavelength 0.1 nm.

2.12 Neutrons are emitted from an atomic pile with a Maxwellian distribution of velocities for temperature 400 K. Find the most probable de Broglie wavelength in the beam.

2.13 A beam of neutrons with energy E runs horizontally into a crystal. The crystal transmits half the neutrons and deflects the other half vertically upwards. After climbing to height H these neutrons are deflected through 90° onto a horizontal path parallel to the originally transmitted beam. The two horizontal beams now move a distance L down the laboratory, one distance H above the other. After going distance L , the lower beam is deflected vertically upwards and is finally deflected into the path of the upper beam such that the two beams are co-spatial as they enter the detector. Given that particles in both the lower and upper beams are in states of well-defined momentum, show that the wavenumbers k, k' of the lower and upper beams are related by

$$k' \simeq k \left(1 - \frac{m_0 g H}{2E} \right). \quad (2.96)$$

In an actual experiment (R. Colella *et al.*, *Phys. Rev. Lett.*, **34**, 1472, 1975) $E = 0.042$ eV and $LH \sim 10^{-3} \text{ m}^2$ (the actual geometry was slightly different). Determine the phase difference between the two beams at the detector. Sketch the intensity in the detector as a function of H .

2.14 A particle moves in the potential $V(\mathbf{x})$ and is known to have energy E_n . (a) Can it have well-defined momentum for some particular $V(\mathbf{x})$? (b) Can the particle simultaneously have well-defined energy and position?

2.15 The states $\{|1\rangle, |2\rangle\}$ form a complete orthonormal set of states for a two-state system. With respect to these basis states the operator σ_y has matrix

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (2.97)$$

¹⁰ In an elegant formulation of quantum mechanics, this last result is the basic postulate of the theory, and one derives other rules for the physical interpretation of the q_n, a_n , etc., from it – see J. von Neumann, *Mathematical Foundations of Quantum Mechanics*, Princeton University Press.

Could σ be an observable? What are its eigenvalues and eigenvectors in the $\{|1\rangle, |2\rangle\}$ basis? Determine the result of operating with σ_y on the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|1\rangle - |2\rangle). \quad (2.98)$$

2.16 A three-state system has a complete orthonormal set of states $|1\rangle, |2\rangle, |3\rangle$. With respect to this basis the operators H and B have matrices

$$H = \hbar\omega \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad B = b \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad (2.99)$$

where ω and b are real constants.

- Are H and B Hermitian?
- Write down the eigenvalues of H and find the eigenvalues of B . Solve for the eigenvectors of both H and B . Explain why neither matrix uniquely specifies its eigenvectors.
- Show that H and B commute. Give a basis of eigenvectors common to H and B .

2.17 Given that A and B are Hermitian operators, show that $i[A, B]$ is a Hermitian operator.

2.18 Given an ordinary function $f(x)$ and an operator R , the operator $f(R)$ is defined to be

$$f(R) = \sum_i f(r_i) |r_i\rangle \langle r_i|, \quad (2.100)$$

where r_i are the eigenvalues of R and $|r_i\rangle$ are the associated eigenkets. Show that when $f(x) = x^2$ this definition implies that $f(R) = RR$, that is, that operating with $f(R)$ is equivalent to applying the operator R twice. What bearing does this result have in the meaning of $e^{R\hbar}$?

2.19 Show that if there is a complete set of mutual eigenkets of the Hermitian operators A and B , then $[A, B] = 0$. Explain the physical significance of this result.

2.20 Given that for any two operators $(AB)^\dagger = B^\dagger A^\dagger$, show that

$$(ABCD)^\dagger = D^\dagger C^\dagger B^\dagger A^\dagger. \quad (2.101)$$

2.21 Prove for any four operators A, B, C, D that

$$[ABC, D] = AB[C, D] + A[B, D]C + [A, D]BC. \quad (2.102)$$

Explain the similarity with the rule for differentiating a product.

2.22 Show that for any three operators A , B and C , the **Jacobi identity** holds:

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0. \quad (2.103)$$

2.23 Show that a classical harmonic oscillator satisfies the virial equation $2\langle KE \rangle = \alpha\langle PE \rangle$ and determine the relevant value of α .

2.24 Given that the wavefunction is $\psi = Ae^{i(kz - \omega t)} + Be^{-i(kz + \omega t)}$, where A and B are constants, show that the probability current density is

$$\mathbf{J} = v(|A|^2 - |B|^2) \hat{\mathbf{z}}, \quad (2.104)$$

where $v = \hbar k/m$. Interpret the result physically.